Practice 2

Objectives

A. Write programs that execute statements repeatedly using nested loops.

Problem 1

1. (15 pts) Write a complete C++ program that prompts the user to enter an integer from **1** to **9** and displays a pyramid, as shown in the following sample run (there are two spaces between the columns, there are no trailing spaces at the end of each row):

```
Enter the number of lines (from 1 to 9): 8
                1
              2 1 2
            3 2 1 2 3
          4 3 2 1 2 3 4
        5 4 3 2 1 2 3 4 5
      6 5 4 3 2 1 2 3 4 5 6
    7 6 5 4 3 2 1 2 3 4 5 6 7
8 7 6 5 4 3 2 1 2 3 4 5 6 7 8
```

Hint: The outer loop should iterate through the rows; the first inner loop should print the appropriate number of leading spaces; the second inner loop should print numbers in descending order; the third inner loop should print numbers in ascending order.

2. (10 pts) Modify the original program to prompt the user to enter an integer from **1** to **15** and display a pyramid, as shown in the following sample run (there are two spaces between **ALL** the columns):

```
Enter the number of lines (from 1 to 15): 13
                              1
                            2 1 2
                          3 2 1 2 3
                        4 3 2 1 2 3 4
                      5 4 3 2 1 2 3 4 5
                    6 5 4 3 2 1 2 3 4 5 6
                  7 6 5 4 3 2 1 2 3 4 5 6 7
                8 7 6 5 4 3 2 1 2 3 4 5 6 7 8
              9 8 7 6 5 4 3 2 1 2 3 4 5 6 7 8 9
            10 9 8 7 6 5 4 3 2 1 2 3 4 5 6 7 8 9 10
          11 10 9 8 7 6 5 4 3 2 1 2 3 4 5 6 7 8 9 10 11
        12 11 10 9 8 7 6 5 4 3 2 1 2 3 4 5 6 7 8 9 10 11 12
13 12 11 10 9 8 7 6 5 4 3 2 1 2 3 4 5 6 7 8 9 10 11 12 13
```